From "Truth Topology" to Readout Bottlenecks: Negative Replication on GSM8K and a Positive Representation–Decoder Dissociation in Procedural Micro-Worlds

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Abstract

We report a three-phase investigation of hidden-state geometry for reasoning correctness and semantic truth status. Phase A tested a global "topology of truth" hypothesis on GSM8K-style traces; this branch failed under stronger controls. Dynamic and fixed-window H_0 summaries were weak or degenerate and did not outperform simple controls after confound checks. Phase B, under fixed decoding, showed a different strong effect: token-cap termination was the dominant operational predictor of error, indicating non-convergence as the primary failure mode. Phase C introduced a procedural micro-world benchmark with nonce lexicons, held-out templates, and exact `True`/`False`/`Unknown` labels. Across Qwen and Gemma families, decoder behavior systematically under-expressed `Unknown`, while verdict-region hidden-state probes recovered substantial `Unknown` signal on held-out worlds. This dissociation survived prompt-path controls, constrained decoding, base-vs-instruct checks, verdict-step label-logit analysis, and layer sweeps. The positive contribution is therefore not a universal scalar "truth topology," but a replicated representation–decoder mismatch for non-entailment.

1 Introduction

This repository started from a broad claim: correct reasoning might have a globally coherent topological signature in hidden-state trajectories. That claim did not survive replication-quality checks. The final evidence supports a narrower and stronger statement:

> On a controlled semantic task, non-entailment (`Unknown`) is encoded in internal representations but systematically under-expressed by decoder outputs.

We report the full trajectory rather than hiding failures:

1. **Phase A (negative):** global trace topology did not yield a reliable correctness signal.
2. **Phase B (positive):** fixed-decoding non-convergence strongly predicted wrong answers.
3. **Phase C (main positive):** replicated representation–decoder dissociation for `Unknown` in a held-out procedural semantic benchmark.

2 Research Questions

RQ1 (failed): Can global topological summaries of natural-language reasoning traces predict correctness robustly across conditions?

RQ2 (supported): In a controlled semantic benchmark, is `Unknown` / non-entailment represented in hidden states even when decoder behavior collapses that class?

3 Phase A: Global Topology on GSM8K (Negative Replication)

3.1 Setup

Phase A used paired Qwen3.5-0.8B/2B pilot traces and computed static/dynamic topological summaries from hidden-state clouds. The main files are in:

- `artifacts/paired_dynamic_qwen35/`
- `artifacts/qwen35_2b_window_corrected/`

3.2 Result

On non-capped 2B pilot traces, dynamic metrics were weak:

- `h0_entropy_final AUC=0.55`
- `peak_h0_entropy AUC=0.55`
- `auc_h0_entropy_per_token AUC=0.50`
- `max_delta_h0 AUC=0.60`

After fixing window preprocessing, topology remained non-competitive:

- controls-only AUC: 0.60
- corrected-topology-only AUC: 0.20
- controls+corrected-topology AUC: 0.20

3.3 Interpretation

The global hypothesis failed because the measured summaries were dominated by construction and length effects, not stable correctness-related geometry. This branch is retained as a negative replication record.

4 Phase B: Fixed-Decoding Convergence Branch

4.1 Setup

We ran Qwen3.5-2B with fixed decoding:

- `temperature=0.7`
- `top_p=0.95`
- `max_new_tokens=384`
- 20 questions $\times$ 9 samples/question ($n = 180$ total)

Primary manifest: `artifacts/qwen35_2b_within_question/run_manifest.csv`

4.2 Outcome and equation

Define

$$C = \mathbb{1}[\text{termination is cap}], \qquad Y = \mathbb{1}[\text{wrong answer}]$$

Counts:

- capped: 87 (75 wrong / 12 correct)
- EOS: 93 (17 wrong / 76 correct)

The operational signal is large:

$$\mathrm{AUC}(C \to Y) = 0.839$$

and the odds ratio is

$$\mathrm{OR} = \frac{\frac{75}{12}}{\frac{17}{76}} \approx 26.40.$$

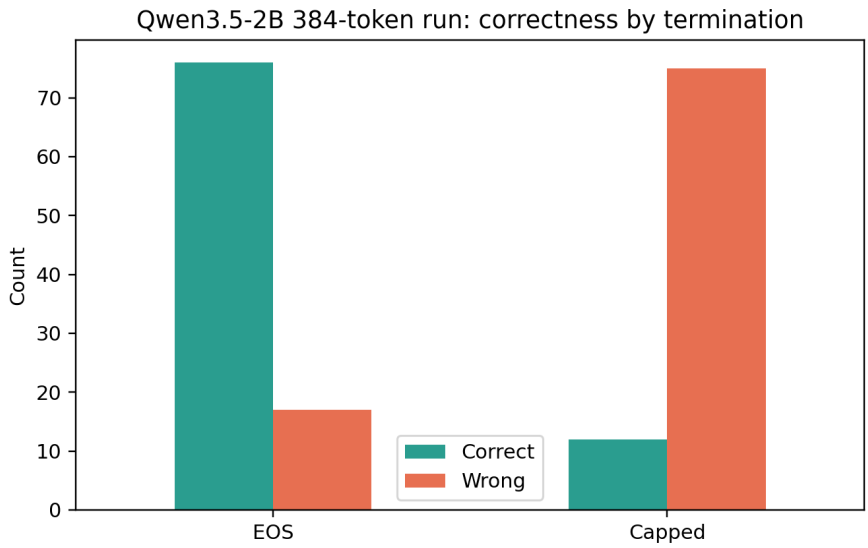

Figure 1: Fixed-config Qwen3.5-2B run: correctness split by termination mode.

4.3 Cap sensitivity at longer budget

We reran all originally capped samples at `max_new_tokens=640` with matched IDs/seeds:

- rerun set: $n = 87$, in `artifacts/qwen35_2b_cap_sensitivity_640/`
- among originally capped wrong (75): 27 flipped wrong→correct, 48 stayed wrong
- still capped at 640 remained heavily wrong (24 wrong / 4 correct)

Conclusion: truncation matters, but deeper non-convergence remains a dominant regime.

5 Phase C: Procedural Micro-World Benchmark

5.1 Dataset design

To isolate semantics from benchmark contamination, we procedurally generate latent worlds and render language statements over nonce symbols. The dataset generator is: `scripts/generate_micro_world_datase` with outputs under `artifacts/micro_world_v1/dataset/`.

Each world has:

- entities and attributes with split-specific nonce lexicon,
- binary relations with held-out lexical variants,
- 9 propositions and 8 paraphrases/proposition,
- exact labels $y \in \{\texttt{True}, \texttt{False}, \texttt{Unknown}\}$.

Thus each world contributes 72 examples.

Main sweeps use 20 train worlds and 20 eval worlds (1,440 eval examples/model).

5.2 Worked world example

Table 1 shows the style of one generated world in human-readable form.

Table 1: Worked micro-world example (illustrative format).

World fact set	Query statement	Gold label
`mep is falm; grel is not falm; nalo foshes sop`	`The object mep has property falm.`	True
`mep is falm; grel is not falm; nalo foshes sop`	`The object grel has property falm.`	False
`mep is falm; grel is not falm; nalo foshes sop`	`The relation fosh holds from mep to nalo.`	Unknown

6 Methods

6.1 Stored inference artifacts

Per example we store prompt text, gold label, decoder label, parse status, verdict-region hidden states, and verdict-step logits. The canonical per-sample files are:

- `sample.json`
- `verdict_states.npz`
- `logits_summary.npz`

6.2 Linear probe

For one state key, with hidden vector $h_i \in \mathbb{R}^d$ and class $y_i \in \{1, 2, 3\}$, the probe is:

$$s_i = Wh_i + b, \quad W \in \mathbb{R}^{3\times d},\ b \in \mathbb{R}^3,$$

$$\hat{y}_i = \arg\max_c (s_i)_c.$$

Training minimizes multiclass cross-entropy:

$$\mathcal{L}(W, b) = -\sum_i \log \frac{\exp((Wh_i + b)_{y_i})}{\sum_c \exp((Wh_i + b)_c)}.$$

This asks whether class information is linearly recoverable from hidden states, without modifying the base model.

6.3 Within-world geometry gap

Within each world and state key, we compute:

$$D_{\text{same}} = \mathbb{E}[d(h_i, h_j) \mid y_i = y_j], \quad D_{\text{diff}} = \mathbb{E}[d(h_i, h_j) \mid y_i \neq y_j],$$

$$\Delta = D_{\text{diff}} - D_{\text{same}}.$$

$\Delta > 0$ implies tighter same-label neighborhoods than different-label neighborhoods. We report sign-test consistency across worlds.

6.4 Verdict-step label-logit metrics

For label logits (ℓ_T, ℓ_F, ℓ_U):

$$P_U = \frac{e^{\ell_U}}{e^{\ell_T} + e^{\ell_F} + e^{\ell_U}}, \qquad m_U = \ell_U - \max(\ell_T, \ell_F).$$

Negative m_U means `Unknown` is under-ranked at the final decision step.

6.5 Layer sweep

For each transformer layer ℓ, we extract the same state family and evaluate the same linear probe protocol:

$$R_U^{(\ell)} = \text{probe recall for Unknown at layer } \ell.$$

This identifies where non-entailment is maximally linearly recoverable.

7 Main Results

7.1 Decoder collapse vs hidden-state recovery

Table 2 compares decoder output and verdict-token probe performance.

Table 2: Decoder vs verdict-token probe ($n = 1440$ eval examples/model).

Model	Decoder Accuracy	Decoder Unknown Recall	Probe Unknown Recall
Qwen3.5-2B	0.436	0.000	0.738
Qwen3.5-4B (no-think)	0.478	0.000	0.129
Gemma-3-4B-it (no-think)	0.397	0.013	0.563

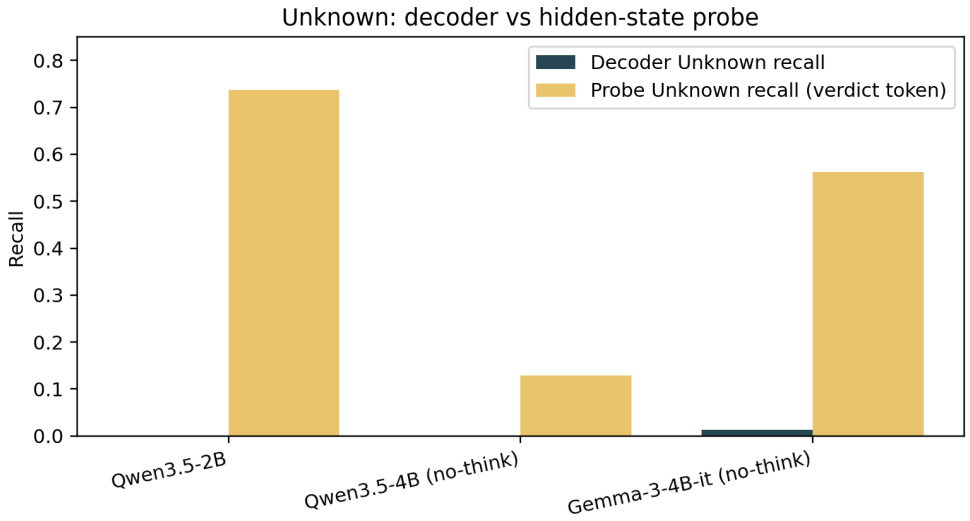

Figure 2: Decoder Unknown recall vs verdict-token probe Unknown recall.

The key effect is replicated: decoder behavior under-expresses `Unknown` while hidden states encode it.

7.2 World-level geometry consistency

Sign tests on world-level geometry gaps were positive across all valid worlds:

- Qwen3.5-2B: 19/19 worlds with $\Delta > 0$,
- Gemma-3-4B-it: 20/20 worlds with $\Delta > 0$.

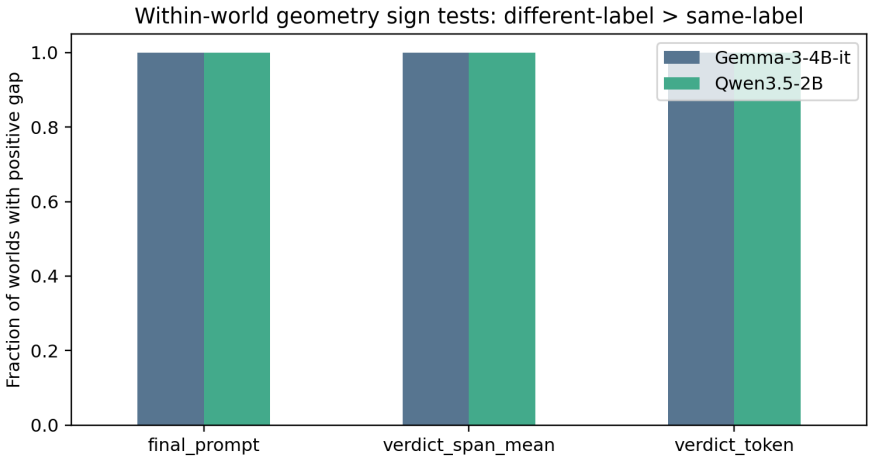

Figure 3: Fraction of worlds with positive same-vs-different label gap.

7.3 Mechanistic controls

The dissociation persisted under:

- raw prompt path control,
- constrained {`True`, `False`, `Unknown`} decoding,
- base-vs-instruct comparison with base-format parse repair,
- verdict-step logit analysis,
- layer-sweep probes.

7.4 Logit evidence

On gold-`Unknown` decoder failures:

- Gemma-3-4B-it raw: mean $P(\texttt{Unknown}) = 0.192$, mean margin $m_U = -1.293$,
- Gemma-3-4B-pt basefmt: mean $P(\texttt{Unknown}) = 0.177$, mean margin $m_U = -1.045$.

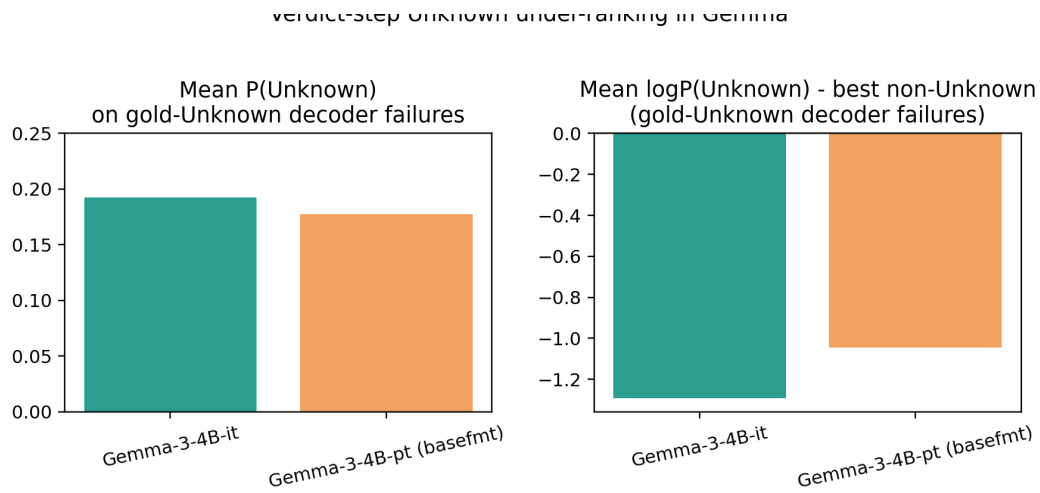

Figure 4: Unknown competitiveness on gold-Unknown decoder failures.

7.5 Layer-sweep evidence

Best Unknown recall by state family:

- Gemma-it: `prompt_last`=0.779 at layer 1; `verdict_token`=0.742 at layer 10
- Gemma-pt-basefmt: `prompt_last`=0.825 at layer 29; `verdict_token`=0.733 at layer 28

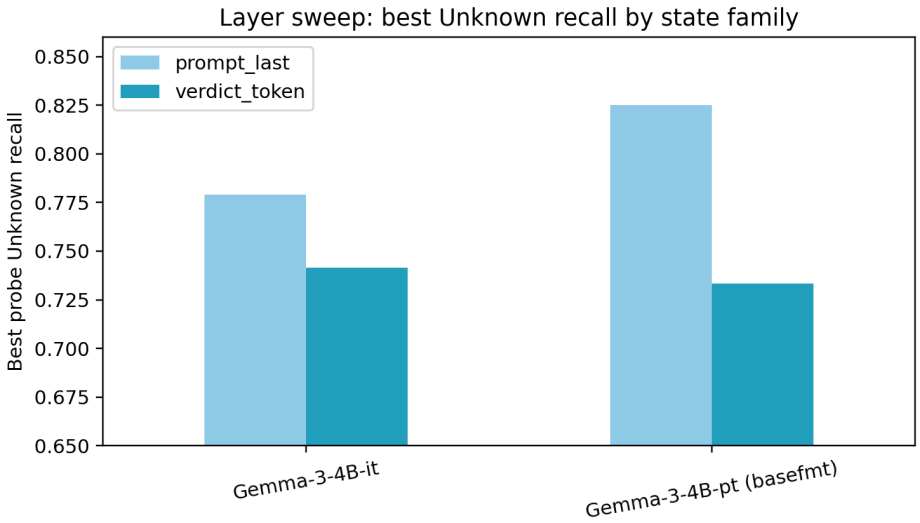

Figure 5: Best Unknown recall from layer sweeps.

8 Interpretation

The evidence supports a representation–readout bottleneck:

1. non-entailment is encoded internally,
2. the decoder under-ranks/under-emits that class,
3. this mismatch replicates across model families and controls.

What is **not** supported:

- a universal scalar "topology of truth,"
- a claim that topology alone is the dominant predictive signal on GSM8K traces.

9 Reproducibility

9.1 Environment

Locked dependency snapshot: `repro/requirements.lock.txt`

9.2 Core commands

From repository root:

```
# Figures
python3 paper/scripts/make_figures.py
```

```
# Compile PDF
cd paper
pdflatex -interaction=nonstopmode -halt-on-error main.tex
pdflatex -interaction=nonstopmode -halt-on-error main.tex
```

9.3 Artifact locations used in this manuscript

- Phase A negative: `artifacts/paired_dynamic_qwen35/`, `artifacts/qwen35_2b_window_corrected/`
- Phase B convergence: `artifacts/qwen35_2b_within_question/`
- Phase B sensitivity: `artifacts/qwen35_2b_cap_sensitivity_640/analysis/`
- Phase C comparisons: `artifacts/micro_world_v1/comparison_*.csv`
- Phase C logits: `artifacts/micro_world_v1/comparison_label_logits_gemma_it_vs_pt_basefmt.csv`
- Phase C layer sweep: `artifacts/micro_world_v1/comparison_layer_sweep_gemma_it_vs_pt_basefmt.csv`

10 Limitations

- Model coverage is small-to-mid scale only.
- The procedural benchmark is controlled but synthetic.
- Probe recoverability does not alone prove causal use by decoder policy.

11 Conclusion

The original global-topology headline did not survive. The validated result is a replicated representation–decoder dissociation: on a held-out procedural semantic benchmark, `Unknown` is strongly recoverable from hidden states while decoder outputs systematically under-express it. This reframes the work from "truth topology" to readout bottlenecks in semantic uncertainty expression.

A Appendix A: Full Reproduction Commands

```
# Phase A dynamic topology
python3 scripts/analyze_paired_dynamic.py \
  --artifact-08b artifacts/paired_qwen35_08b_10 \
  --artifact-2b artifacts/paired_qwen35_2b_10 \
  --out-dir artifacts/paired_dynamic_qwen35
```

```
# Phase A corrected window test
python3 scripts/analyze_2b_window_corrected.py \
  --artifact-dir artifacts/paired_qwen35_2b_10 \
  --out-dir artifacts/qwen35_2b_window_corrected

# Phase B fixed-decoding run
python3 scripts/run_multi_sample_questions.py \
  --model-id Qwen/Qwen3.5-2B \
  --question-start 0 --question-count 20 --samples-per-question 9 \
  --temperature 0.7 --top-p 0.95 --max-new-tokens 384 \
  --artifact-dir artifacts/qwen35_2b_within_question

# Phase B cap sensitivity
python3 scripts/run_cap_sensitivity.py \
  --model-id Qwen/Qwen3.5-2B \
  --source-artifact-dir artifacts/qwen35_2b_within_question \
  --out-dir artifacts/qwen35_2b_cap_sensitivity_640 \
  --temperature 0.7 --top-p 0.95 --max-new-tokens 640

# Phase C dataset
python3 scripts/generate_micro_world_dataset.py \
  --out-dir artifacts/micro_world_v1/dataset \
  --seed 1729 --train-worlds 100 --dev-worlds 25 --eval-worlds 100 \
  --props-per-world 9 --paraphrases-per-prop 8
```